# MATH 345 Skeleton Notes
# Unit 3: Beyond linear maps: local linearization
# Section 3.5: Local linearization and differentiability

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## Tangent planes and scalar local linearization

### Linear approximations and differentiability

Recall that the linear approximation of a function $f : \mathbb{R} \to \mathbb{R}$ of one variable at some point $a$ is the affine function $L(x) = f(a) + f'(a)(x - a)$, and this function gives the best local affine approximation of $f$ near $a$. Similarly, the partial derivatives of a function can be used to define a local approximation to a function of several variables. This leads to Linear Approximation.

**Definition: Linear Approximation** Given a scalar-valued function $f$, and a point $\mathbf{a}$ in the domain of $f$ where the partial derivatives of $f$ are well-defined, the $\boxed{\text{linear approximation}}$ of $f$ at the point $\mathbf{a}$ is the affine function

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a})^T(\mathbf{x} - \mathbf{a}),$$

where we can expand the dot product that occurs in the equation, so that

$$L(\mathbf{x}) = f(\mathbf{a}) + \sum_{i=1}^{n} (D_i f)(\mathbf{a})(x_i - a_i).$$

The approximation is affine in $\mathbf{x}$; its linear part is the map $\mathbf{h} \mapsto (\nabla f)(\mathbf{a})^T \mathbf{h}$ on small input changes.

In particular, the linear approximation of a scalar-valued function $f(x, y)$ at a point $(a, b)$ is given by

$$L(x, y) = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b).$$

**Activity**

Consider the function

$$f(x, y, z) = x^2y + yz^2.$$

Find the linear approximation of the function

$$f(x, y, z) = x^2y + yz^2$$

at the point $(1, 2, 0)$.

**Activity**

Consider the function

$$f(x, y, z) = x^2y + yz^2.$$

Use the linear approximation you have found to approximate the value $f(1.04, 1.97, 0.05)$.

Now suppose a function $f$ of a single variable is differentiable at a point

$a$, i.e., so that the limit

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

exists. We can rearrange this equation to read

$$\lim_{x \to a} \frac{f(x) - f(a) - f'(a)(x - a)}{x - a} = 0,$$

i.e., so that

$$\lim_{x \to a} \frac{f(x) - L(x)}{x - a} = 0,$$

where $L(x) = f(a) + f'(a)(x - a)$ is the linear approximation of $f$ at $a$. A function of several variables is *differentiable* when the same property holds, with the linear approximation given by the partial derivatives of the function.

**Definition: Differentiability** A scalar-valued function $f$ is differentiable at a point $\mathbf{a}$ if the gradient of $f$ is well-defined at $\mathbf{a}$, and if

$$\lim_{\mathbf{x} \to \mathbf{a}} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|} = 0,$$

where

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a})^T(\mathbf{x} - \mathbf{a})$$

is the linear approximation of $f$ at $\mathbf{a}$. A vector-valued function is differentiable at a point precisely when each of its component functions is differentiable at that point.

**Note** The equation

$$\lim_{\mathbf{x}\to\mathbf{a}} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|} = 0$$

can only hold if $L$ is a *good* linear approximation of $f$ near $\mathbf{a}$, so that the magnitude of the numerator is much smaller than the magnitude of the denominator when $\mathbf{x}$ is near $\mathbf{a}$, i.e., when $\|\mathbf{x} - \mathbf{a}\|$ is small.

The following theorem is the main way we will determine if a function is differentiable.

**Theorem: Continuity of First Partials Implies Differentiability**
Let $f$ be a scalar-valued function, and suppose $\mathbf{a}$ is an interior point (recall Interior, boundary, and closed sets) of the domains of $f$ and its partial derivatives. Moreover, suppose that all of the partial derivatives $D_1 f, \ldots, D_n f$ are continuous at $\mathbf{a}$. Then $f$ is differentiable at $\mathbf{a}$.

## Tangent planes

**Activity**

Find an equation for the plane containing the lines $L_1$ and $L_2$, where $L_1$ is the line parameterized by the function

$$\mathbf{r}(t) = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

and where $L_2$ is the line parameterized by the function

$$\mathbf{r}(s) = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + s \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}.$$

**Definition: Tangent Plane** Let $S$ be a surface, and $\mathbf{x}_0$ a point on $S$. A **tangent plane** to $S$ at $\mathbf{x}_0$ is a plane that contains the tangent lines of any differentiable curve contained in $S$ as it passes through $\mathbf{x}_0$. A tangent plane touching a surface along tangent lines through a point. A curved orange surface $S$ is shown with a translucent blue plane touching it at the labeled point $P_0(x_0, y_0, z_0)$. Several colored curves pass through $P_0$ on the surface, and straight tangent lines through the same point lie in the plane. The drawing emphasizes that all tangent directions to curves through $P_0$ sit inside the tangent plane.
The tangent plane to a surface $S$ at a point $P_0$ contains all the tangent lines to curves in $S$ that pass through $P_0$. Figure 4.27 from Edwin "Jed" Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 4.27.

**Theorem: Equation of the Tangent Plane** Suppose $S$ is the graph of a function $f(x, y)$ which is differentiable at a point $(x_0, y_0)$. Then the tangent plane to $S$ exists, is unique, and is described by the equation

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Note that this plane is the graph of the linear approximation $L$ to $f$ at $(x_0, y_0)$.

**Proof** We verify that the plane described by the equation above is the only possible tangent plane. Recall from Geometric Interpretation that the curve obtained by the intersection of $S$ with the plane $y = y_0$ has slope $f_x(x_0, y_0)$ at the point $(x_0, y_0)$, and the tangent line to the curve at this point is by a line with direction vector $(1, 0, f_x(x_0, y_0))$. Similarly, the curve obtained by the intersection of $S$ with the plane $x = x_0$ has slope $f_y(x_0, y_0)$, and so the tangent line to the curve at this point is a line with direction vector $(0, 1, f_y(x_0, y_0))$. But now using the method of the activity, we see that a normal vector to the plane containing these two lines is of the form $\mathbf{n} = (-f_x(x_0, y_0), -f_y(x_0, y_0), 1)$. So the tangent plane to $S$ at $(x_0, y_0, f(x_0, y_0))$ must be described by the equation

$$(-f_x(x_0, y_0), -f_y(x_0, y_0), 1) \cdot (x - x_0, y - y_0, z - f(x_0, y_0)) = 0,$$

an equation we may rearrange to read

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

**Activity**

Find an equation of the tangent plane to the surface

$$z = e^x \cos y$$

at $(0, \frac{\pi}{2}, 0)$.

# Vector-valued local linearization

## Linear maps reappear locally

A nonlinear map is usually not linear globally. Near one input, however, differentiability replaces small input changes by a linear map.

$$F(\mathbf{a}+\mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

$\mathbf{x} \mapsto F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$ is affine in $\mathbf{x}$. $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ is the linear map on input changes.

Linear algebra gave us global linear maps $\mathbf{x} \mapsto A\mathbf{x}$. Calculus gives us local linear maps $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$. The same matrix language therefore reappears inside nonlinear problems.

**Definition: Linear Approximation of Vector-Valued Functions**

Let $F : \mathbb{R}^n \to \mathbb{R}^m$ be a vector-valued function. Provided that the partial derivatives of the components of $F$ all exist at $\mathbf{a}$, so that the Jacobian matrix $J_F$ is well-defined, the

linear approximation of $F$ at $\mathbf{a}$ is the vector-valued affine function

$$L(\mathbf{x}) = F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$$

where $J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$ is the Jacobian matrix of $F$ evaluated at $\mathbf{a}$, and then applied to the vector $\mathbf{x}-\mathbf{a}$. The approximation is affine in $\mathbf{x}$; its linear part is the map $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ on small input changes.

The components of $L$ are precisely the linear approximations of the components of $F$, i.e., $L_i(\mathbf{x}) = F_i(\mathbf{a}) + (\nabla F_i)(\mathbf{a})^T(\mathbf{x}-\mathbf{a})$.

**Note** Geometrically, the graph of the linear approximation to a vector-valued function can be visualized as a higher dimensional “tangent plane” of the graph of the function $F$ at $\mathbf{a}$.

**Activity**

Let $F(x, y) = (x^2y, xe^y)$. Find the linear approximation of $F$ at the point $(1, 0)$, and use it to approximate $F(1.1, -0.2)$.

**Activity: Local square-grid visualization**

Return to

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

The Jacobian matrix is

$$J_F(x, y) = \begin{bmatrix} 1 & 2y \\ 0 & 1 \end{bmatrix}.$$

Let

$$\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}.$$

1. Compute $J_F(\mathbf{a})$.
2. Let $\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$. Compute $F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a})$.
3. Compute $J_F(\mathbf{a})\mathbf{h}$.
4. Compute the error

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - J_F(\mathbf{a})\mathbf{h}.$$

5. If $0 \leq h_2 \leq r$, how large can the first coordinate of the error be?
6. What happens to the local square-grid picture as $r$ gets smaller?

**Note** The local square-grid visualization does not produce one matrix for the whole nonlinear map. It shows one matrix approximation near one base point.

The approximation in this unit is local. Near one input $\mathbf{a}$, a differentiable function behaves like

$$F(\mathbf{a}+\mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

Unit 4 studies a different kind of approximation:

$$\min_x \|Ax - b\|.$$

That is a global best approximation by a vector in a subspace. Unit 3 asks what a nonlinear function looks like near one point; Unit 4 asks which allowable vector is closest overall.